

# Energy Management System of Microgrid using Optimization Approach

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**Abstract:** – Microgrid technology can efficiently integrate a new practical way for large-scale application of grid-connected generation of renewable energy. An Energy Management System (EMS) in microgrid, is important for optimum use of the distributed energy resources in smart, protected, consistent, and synchronized ways. This paper discusses the management of Energy Storage System (ESS) connected in a microgrid with a solar array and control the battery discharge and charge operations with converter based on the necessity of the demand. Problem for one-day energy management of microgrid is discussed. This paper focuses on analyzing of heuristic and optimization approach for minimizing total variable electricity prize for clear and cloudy day. The output variables like power of PV, grid, ESS, and loads, grid voltage, ESS state of charge and price graphs are analyzed for each case. The MATLAB software is used for simulating this case study.

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**Keywords:** Microgrid, Energy Storage System, Solar Array, Linear Programming, Energy Management System.

## Abbreviation

$V_{bat}(T)$  = Terminal voltage under no-load condition of battery

$Q_{bat}(T)$  = battery capacity (Ah)

$I_{bat}(T)$  = current of battery (A)

$E_{bat}^{rated}$  = rated capacity of the battery (Ah)

$E_{bat}$  = energy stored in battery (Ah)

$P_{bat}$  = power from battery (MW)

$m$  = polarization voltage (V)

$P_{pv}$  = total generation of solar array (MW),

$P_{grid}$  = power from grid (MW)

$V_o(T)$  = constant voltage of battery (V)

$P_{load}$  = total load demand

$A$  = exponential voltage (V)

$C_{tot}$  = variable electricity price

$C_{grid}$  = grid cost

$E_{grid}$  = energy stored in grid

$k$  = time step.

$B$  = exponential battery capacity (Ah–1)

## 1. INTRODUCTION

Microgrids (MGs) are presently receiving great attention and are considered the future trend for power distribution systems [1]. In a microgrid, it is necessary to maintain the power balance for stability because of the uncertain generation of the renewable energy sources (RESs) [2]. The reliability of the RESs are an important task because of mismatch between supply and energy demand. Energy Storage Systems (ESSs) are an important part of the MG, because they allow for a better utilization of the RESs. The Energy Management System (EMS) in an MG is defined as a control system that offers the essential functionality, which guarantees that both the distribution and generation systems deliver power at minimum

operating costs. The EMS plays a significant role in increasing the reliability and efficiency of an MG.

Many researchers have solved these energy management strategies using various solution approaches to achieve the optimal and efficient operation of the MG. Sukumar et al. [3] proposed an EMS based on nonlinear and linear programming methods, Heymann et al. [4] proposed an EMS based on rule-based and dynamic programming methods, Elsieid et al. [5] developed a multi-objective EMS based on swarm and genetic optimization, Tiwari and Srivastava [6] proposed an EMS based on other meta-heuristic approaches, ArcosvAviles et al. [7] presented EMS based on neural network and fuzzy logic, Ghorbani Rahmani, and Unland [8] developed an EMS based on multi-agent system, Liu et al. [9] proposed EMS based on other artificial intelligent methods, Ghasemi [10] proposed an EMS based on stochastic and robust programming approaches.

In this work we have used two blocks, apart from variable load and fixed load, a solar PV array block which represents as an RES and another ESS block which act as a storage element for the system. Because of solar irradiance we have made a selection of clear day and cloudy day. This model walks through the process of developing an optimization routine that uses forecast pricing and loading conditions to optimally store/sell energy from a grid-scale battery system. This real-time data is driven for one day. Battery controller is used for storage of energy in battery. When there is surplus power in the MG it stores the surplus power, and provides supplementary power if there is a power deficiency in the MG. The control strategy assumes that the MG does not depend completely on the power delivered by the main grid, and the power supplied by the solar power generation and storage are sufficient at most of times. This study is focused on two approaches: Heuristic approach using State Machine Logic (State flow method) and optimization approach using linear programming method. Linear programming method is done using Linprog command.

The objective of this work is to minimize variable electricity price of MG. The remaining paper is structured as follows: The stochastic modelling of solar and battery is discussed in Section 2 and Section 3, respectively. The formulation of EMS in the presence of solar and battery is discussed in Section 4. Furthermore, the analysis and discussion of various cases are presented in Section 5, followed by concluding remarks in Section 6.

## 2. SOLAR MODELLING

Several types of solar cell materials are being used worldwide but nowadays, silicon is ascendant material used in solar cells because of its impetus, adaptability and absorption of light efficacious. This Photovoltaic array has a total of 24 modules per array comprised of 2 modules per string connected in parallel and 12 strings in series. The 72 Photovoltaic cells are connected in series to get 535kW solar panel. Figure 1 shows the block diagram of the Solar panel.

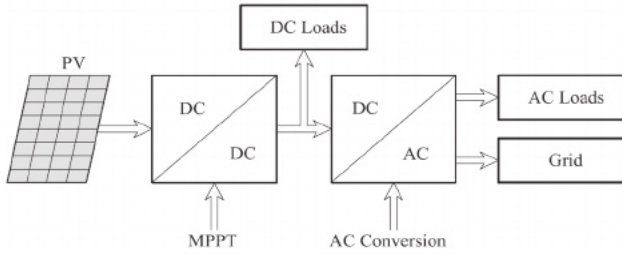


Figure 1. Block Diagram of Solar Panel.

By using DC-DC converter the output power of Photovoltaic array can be controlled by Maximum Power Point Tracker (MPPT) in real time. The Maximum Power Point (MPP) is dependent on shifts in solar radiation and room temperature. In Incremental Conductance (INC) algorithm, the decided gradual change of the solar panel conduction is measured and then differentiate the instant conductance, which is well-known as the ratio of  $I/V$ . After being compared, in reference to the MPP, the position of the current( $I$ ) and voltage( $V$ ) changes [11].

## 3. BATTERY MODELLING

Here, distributed battery with capacity of 3000kWh model is established based on the MATLAB. The current state-of charge ( $PSOC_{bat}(T)$ ) is calculated from the initial state of charge ( $SOC_{ini}(T)$ ) of the battery. The battery voltage is calculated by

$$V_{bat}(T) = V_o(T) - \frac{m \cdot Q_{bat}(T)}{Q_{bat}(T) - \int I_{bat}(T) dt} + A \exp(-B \int I_{bat}(T) dt) \quad (1)$$

The  $PSOC_{bat}(T)$  is calculated by

$$PSOC_{bat}(T) = SOC_{ini}(T) \pm \frac{I_{bat}(T) \cdot \Delta T}{Q_{bat} \cdot 3600} \quad (2)$$

where,  $\Delta T$  is the changing time and  $\pm$  indicate battery charging or discharging, respectively. Available energy to supply at any time instant can be calculated by

$$E_{bat}^{avil}(T) = Q_{bat}(T) V_{bat}(T) PSOC_{bat}(T) \quad (3)$$

The essential energy for storage into the battery is calculated by

$$E_{bat}^{req}(T) = E_{bat}^{rat} - E_{bat}^{avil}(T) \quad (4)$$

## 4. PROBLEM FORMULATION

The main objective is to minimize the total variable electricity price which includes the cost of grid, energy stored while satisfying all operational constraints. Therefore, the objective function can be formulated as

$$C_{tot} = \sum_{k=0}^N C_{grid}(k) * E_{grid}(k) \quad (5)$$

The decision variables for this problem are listed below:

$$x = [P_{grid}(1:N) \text{ } att(1:N) \text{ } E_{batt}(1:N)]^T \quad (6)$$

where,  $P_{grid}(1:N)$  is the power from grid used from time step 1 to  $N$ ,  $P_{batt}(1:N)$  is the power from battery, and  $E_{batt}(1:N)$  is the energy stored in battery.

### 4.1 Constraints

The power input/output of battery is defined as

$$E_{batt}(k) = E_{batt}(k-1) + P_{batt}(k) \Delta T \quad (7)$$

where,  $\Delta T$  is the different time step.

The power balance equation for all the buses are considered as equality constraints, which are defined as

$$P_{pv}(K) + P_{grid}(K) + P_{batt}(K) = P_{load}(K) \quad (8)$$

The second one is related to decision variables as

$$P_{gi}^{min} \leq P_{gi} \leq P_{gi}^{max} \quad (9)$$

$$E^{min} \leq E \leq E^{max} \quad (10)$$

where,  $P$  is the active power of generator in MW and  $E$  is the energy of battery in Ah.

## 5. SIMULATION RESULTS

Figure 2 shows the test system. In the test system, the utility grid is present having capacity of 13.8 kV. Utility grid is connected with three-phase two-winding transformer. The rating of primary winding of three-phase transformer is 13.8 kV and secondary winding is 5 kV and it has star-star connection. Next, a three-phase circuit breaker is connected which follows a closed path for supply of current. After that, the three-phase circuit breaker is connected with variable load to supply the power. The capacity of variable load is 320 kW and another one fixed load is present and its capacity is 350 kW. So, the available total load of the model is 670 kW. Next block represents solar array and its capacity is 535 kW and another one is ESS, which can be known as heart of the system. The rated power of ESS is 400 kW and capacity are 3000 kWh. In this system, we are using two different conditions: clear day and cloudy day. Using these different conditions, we find the total cost of electricity.

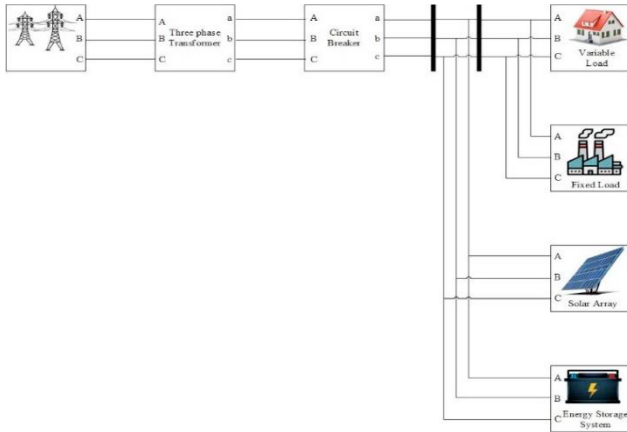


Figure 2. Microgrid test system.

Figure 3 shows the simulation waveforms of clear day analysis using heuristic approach. The first waveform represents the microgrid RMS voltage (Vrms) which shows the fluctuation in voltages because of change in load demands and generations. The second waveform represents the power of Photovoltaic (PV), grid, battery and load. The third waveform represents the ESS State of Charge (SOC) and fourth waveform represents the price of electricity. Here, at the interval of 0 to  $2 \times 10^4$  seconds, the load demand is less than 500kW, so, during that time the ESS State of Charge (SOC) represents its initial capacity i.e., 50%. So, corresponding to that the operating cost is minimum. In the interval of 2 to  $4 \times 10^4$  seconds, the load is increased up to 500 kW and the PV power starts to increase while battery power falls in negative. Because of the increase in the power of PV above the load, surplus power is generated which gets stored in the battery. This means the battery slowly starts to charge. Again, as the cost of renewable generation is negligible, the cost remains low. In the interval of 4 to  $6 \times 10^4$  seconds, the load remains constant and when the battery voltage reaches the nominal battery voltage, it reaches the charging mode. As a result, SOC has a charging performance index, estimated over 80%. Correspondingly, the PV power gets reduced and the battery slowly starts to supply the power which means the battery starts discharging. Herewith, the cost still remains low as we use the power stored in battery because of renewable source. In the interval of 6 to  $8 \times 10^4$  seconds, the load is increased up to 500kW but still the power from the battery though the PV power goes on decreasing. At an instant the battery gets fully discharged and the PV power is negligible, then comes the grid power. The power required for the load is supplied using grid power which raises the cost to maximum. The total cost of a clear day using heuristic approach is \$662.3.

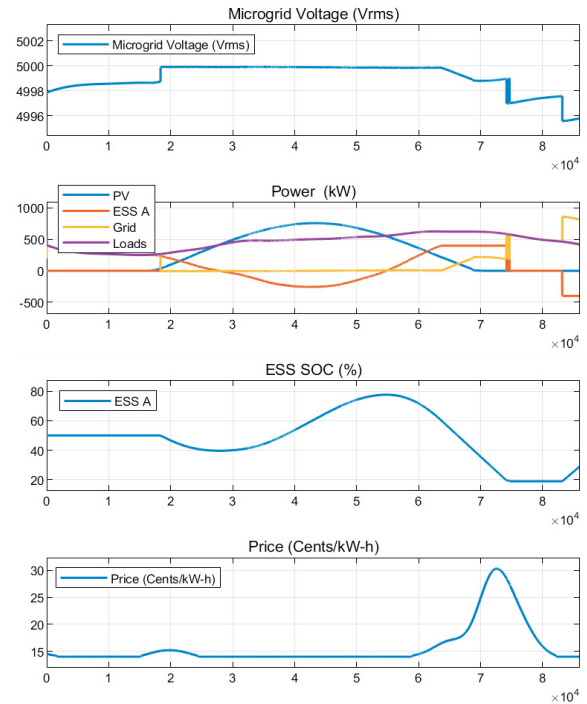


Figure 3. Clear day simulation result using Heuristic Approach.

Figure 4 shows the simulation waveforms of clear day analysis using optimization approach. Here, at the interval of 0 to  $2 \times 10^4$  seconds, the load demand is less, so, during that time the solar array and battery both do not supply power. The grid supplies power and during that time period battery gets fully charged. Afterwards, grid supply is decreasing because of solar array and battery supply the power and battery gets discharged. Correspondingly, the operating cost is minimum. In the interval of 2 to  $4 \times 10^4$  seconds, the load is slightly increasing up to 500kW. The PV power is on the verge of increasing. Simultaneously, battery power is decreasing.

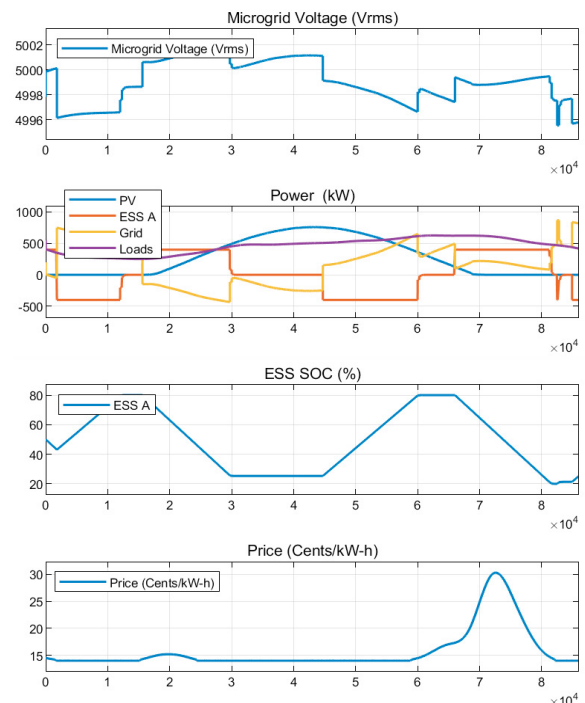


Figure 4. Clear day simulation result using Optimization Approach.

Here, PV supply more power and the grid power decreases which means the battery does not supply. Again, as the cost or renewable generation is negligible the cost remains low. In the interval of  $4$  to  $6 \times 10^4$  seconds, the load remains constant and when the battery voltage reaches the nominal battery voltage, it reaches the charging mode. As a result, SOC has a charging performance index, estimated over 80%. However, the cost still remains low as we use the power stored in battery because of renewable source. In the interval of  $6$  to  $8 \times 10^4$  seconds, the load is increased up to 500kW but still we get the power from the battery though the solar array power goes on decreasing. At an instant the battery slowly discharges and the solar power is negligible, then comes the grid power. The power required for the load is supplied using grid power which raises the cost to maximum. The total cost of a clear day using optimization approach is \$618.7.

Figure 5 shows the simulation waveforms of cloudy day analysis using heuristic approach. Here, at the interval of  $0$  to  $2 \times 10^4$  seconds, as the load demand is less, so, the ESS SOC represents its initial capacity i.e., 50% and corresponding to that the operating cost is minimum. In the interval of  $2$  to  $4 \times 10^4$  seconds, the load is slightly increasing up to 500 kW. The solar array power starts increasing while battery simultaneously fluctuates between that time period because of change in weather. Because of increasing in the solar array above the load, surplus power is generated which gets stored in the battery which means the battery starts to charge. Again, as the cost or renewable generation is negligible the cost remains low. In the interval of  $4$  to  $6 \times 10^4$  seconds, the load remains stable but the battery is not fully charged due to changes in the weather. Accordingly, the solar power decreases and the ESS slowly increases and the battery slowly starts to supply the power. After an instant, solar array increase above the load and surplus power gets stored in the battery.

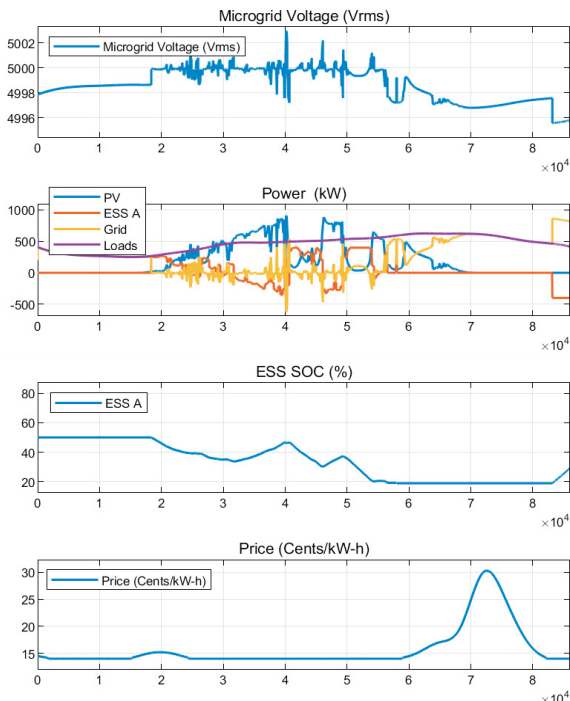


Figure 5. Cloudy day simulation result using Heuristic Approach.

Again, the cost still remains low. In the interval of  $6$  to  $8 \times 10^4$  seconds, the load is increased up to 500kW, at an instant the battery gets discharged and the solar power is negligible, then comes the grid power. The power required for the load is supplied using the grid power which raises the cost to maximum. The total cost of a cloudy day using heuristic approach is \$1052.

Figure 6 shows the simulation waveforms of cloudy day analysis using optimization approach. Here, at the interval of  $0$  to  $2 \times 10^4$  seconds, the load demand is less, so, during that time the solar array and battery both do not supply power. The grid supplies power and in that time period battery is charging to its full capacity. After an instant, grid supply decreases because of the power supply from solar array and battery. Accordingly, the operating cost is minimum. In the interval of  $2$  to  $4 \times 10^4$  seconds, the load is slightly increasing up to 500kW. The power of solar array increases while battery supplies power and reach its nominal capacity. Again, as the cost or renewable generation is negligible the cost remains low. In the interval of  $4$  to  $6 \times 10^4$  seconds, the load remains constant and when the battery voltage reaches the nominal battery voltage, it reaches the charging mode. As a result, SOC has a charging performance index estimated over 80%. Again, the cost still remains low. In the interval of  $6$  to  $8 \times 10^4$  seconds, the load is increased up to 500kW but still we get the power from the battery though the solar array power goes on decreasing. At an instant the battery gets slowly discharging and the solar power is negligible, then comes the grid power. The power required for the load is supplied using the grid power which raises the cost to maximum. The total cost of a cloudy day using optimization approach is \$907.

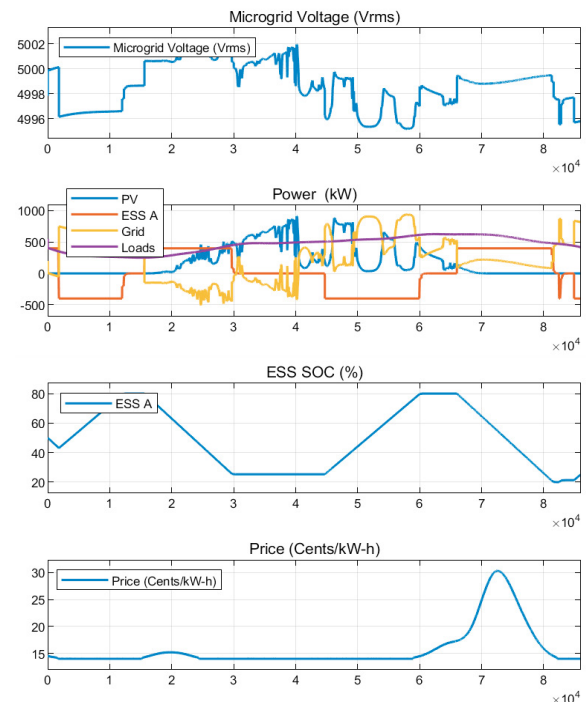


Figure 6. Cloudy day simulation result using Optimization Approach.

Figure 7 is representing cost comparison of grid cost and grid usage for 24 hours. The first waveform represents the grid cost and second waveform represents the grid usage. In the beginning, the use of grid power is low because of the supply

of power from solar array and battery. So, during that time period our grid cost is low. Afterwards, solar power varies due to change in weather condition, so the power is supplied from the grid. Accordingly, the price also varies. At the end, power demand is high and grid price is also high.

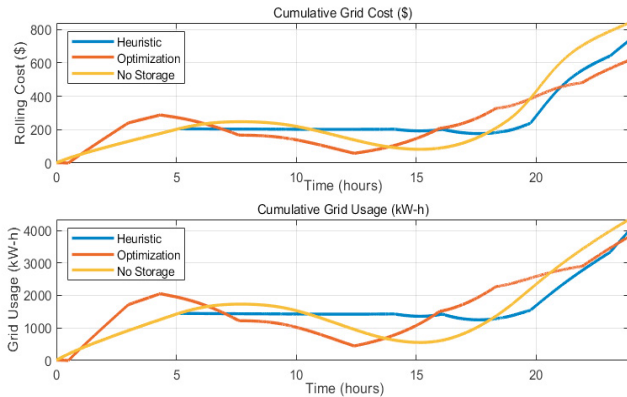


Figure 7. Cost comparison of grid cost and grid usage for 24 hours.

## 6. CONCLUSIONS

This paper represents the integrated Energy Management System (EMS) model of Microgrid (MG). EMS is an important issue owing to its significance in the safe and inexpensive operation of the load. The objective of this study is to minimize the variable electricity price of MG. The proposed model is performed on MATLAB environment. Based on the literature survey, different authors have addressed the problem and provided solutions. These methods are chosen based on their practicality, reliability, and resource availability in the microgrid environment. In this paper two methods are used: the heuristic and optimization methods. The cost comparison of heuristic method is higher for clear day with 11.17% and cloudy day with 14.80% compared to optimization method. Therefore, the results indicate that the optimization method consumes less cost which act as an advantage of the method.

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